

pharmaniaga®

Suxamethonium Chloride

100mg/2mL Solution for Injection



COMPOSITION

Each 2mL ampoule contains 100mg of suxamethonium chloride.
Excipients: Hydrochloric Acid q.s. to pH 3.0–4.5 and Water for Injection q.s. to 2mL.

DESCRIPTION

A colourless solution in 2 mL amber glass ampoules.
Pharmaniaga Suxamethonium Chloride 100mg/2mL Solution for Injection is compatible with the following infusion solutions:

- 0.9% Sodium Chloride solution.
- 5% Dextrose solution.

The solution should be colourless solution when diluted. The solution will remain stable for 24 hours at both temperature 2-8°C and below 30°C after diluted with the compatible solutions listed above.

PHARMACODYNAMICS

Pharmacotherapeutic group: Peripherally acting muscle relaxants, choline derivatives, ATC code: M03AB01.
Short-acting depolarising neuromuscular blocking agent.
Suxamethonium, an analogue of acetylcholine, inhibits neuromuscular transmission by depolarising the motor end plates in skeletal muscle. The depolarisation may be observed as fasciculation. Subsequent neuromuscular transmission is inhibited as long as an adequate concentration of suxamethonium remains at the receptor site. Onset of flaccid paralysis occurs within 30-60 seconds of intravenous injection and with single administration persists for 2-6 minutes. The paralysis following administration of suxamethonium is progressive, with differing sensitivities of different muscles. This initially involves consecutively the levator muscles of the face, muscles of the glottis and finally the intercostals and the diaphragm and all other skeletal muscles. The short duration of suxamethonium is considered to be due to its rapid metabolism in the blood. Suxamethonium is rapidly hydrolysed by plasma cholinesterase to succinylcholine and then more slowly to succinic acid and choline.

PHARMACOKINETICS

Absorption:

Following intravenous administration, there is rapid hydrolysis by pseudocholinesterase with the initial metabolite being succinylmonocholine a weak neuro-muscular drug.

Distribution:

The pharmacokinetic parameters for suxamethonium 1mg/kg and 2mg/kg respectively are: Apparent volume of distribution 16.4 ± 14.7mL/kg and 5.6 ± 6.8mL/kg; area under the plasma concentration-time curve 124.3 ± 163.2 and 695 ±1008.9 minutes/microgram/ml.
The arterial blood suxamethonium concentration at 30 and 120 seconds following injection of suxamethonium 1mg/kg are 79.5 ± 108.4 and 3.3 ± 6.7 micrograms/ml, and following injection of 2mg/kg suxamethonium are 336.2 ± 512.5 and 7.2 ±13.0 micrograms/ml, respectively.
Only a small fraction of suxamethonium reaches the neuromuscular junction. Its action is terminated by diffusion away from the end plate. Succinylcholine does not readily cross the placenta.

Elimination:

Succinylmonocholine is metabolised to succinic acid with only a small amount excreted in the urine. The pharmacokinetic parameters for suxamethonium 1mg/kg and 2mg/kg respectively are: total body clearance 40.5 ± 38.7 and 15.0 ± 14.8 litres/min and elimination half-life 16.6 ± 4.8 and 11.7 ± 4.5 seconds.

INDICATIONS

Suxamethonium is used in anaesthesia as a muscle relaxant to facilitate endotracheal intubation, mechanical ventilation and a wide range of surgical and obstetric procedures. It is also used to reduce the intensity of muscular contractions associated with pharmacologically or electrically-induced convulsions.

RECOMMENDED DOSAGE

Usually by intravenous injection.

Adults

The dose is dependent on body weight, the degree of muscular relaxation required, the route of administration, and the response of individual patients.
To achieve endotracheal intubation Suxamethonium is usually administered intravenously in a dose of 1 mg/kg. This dose will usually produce muscular relaxation in about 30 to 60 seconds and has a duration of action of about 2 to 6 minutes. Larger doses will produce more prolonged muscular relaxation, but doubling the dose does not necessarily double the duration of relaxation. Supplementary doses of Suxamethonium of 50% to 100% of the initial dose administered at 5 to 10 minute intervals will maintain muscle relaxation during short surgical procedures performed under general anaesthesia.
For prolonged surgical procedures Suxamethonium may be given by intravenous infusion as a 0.1% to 0.2% solution, diluted in 5% glucose solution or sterile isotonic saline solution, at a rate of 2.5 to 4 mg per minute. The infusion rate should be adjusted according to the response of individual patients.
The total dose of Suxamethonium given by repeated intravenous injection or continuous infusion should not exceed 500 mg per hour.

Children

Infants and young children are more resistant to Suxamethonium compared with adults.
The recommended intravenous dose of Suxamethonium for neonates and infants is 2 mg/kg. A dose of 1 mg/kg in older children is recommended.
When Suxamethonium is given as intravenous infusion in children, the dosage is as for adults with a proportionately lower initial infusion rate based on body weight. Suxamethonium may be given intramuscularly to infants at doses up to 4 to 5mg/kg and in older children up to 4 mg/kg. These doses produce muscular relaxation within about 3 minutes. A total dose of 150 mg should not be exceeded.

Use in the elderly

Dosage requirements of Suxamethonium in the elderly are comparable to those for younger adults.
The elderly may be more susceptible to cardiac arrhythmias, especially if digitalis-like drugs are also being taken.

ROUTE OF ADMINISTRATION

Parenteral.

CONTRAINDICATIONS

Hypersensitivity to the active substance or to any of the excipients listed.
Suxamethonium should not be administered to a patient who is not fully anaesthetised. Neuromuscular function should be monitored when suxamethonium is being used over a prolonged period.
Suxamethonium causes a significant transient rise in intra-ocular pressure and therefore it should not be used in the presence of glaucoma, detached retina or open eye injury unless the potential benefit of its use outweighs the potential risk to the eye.
Suxamethonium is contra-indicated in patients with a personal or family history of malignant hyperthermia.
Suxamethonium can trigger sustained myofibrillar contractions in susceptible individuals. If this occurs, all anaesthetic agents which are known to be associated with it (including suxamethonium) must be stopped and full supportive measures implemented immediately. Intravenous dantrolene sodium is the primary specific therapeutic drug and should be given as soon as possible after the diagnosis is made.
Suxamethonium is contra-indicated in patients with inherited atypical or low serum level of pseudocholinesterase.

Prolonged and intensified neuromuscular blockade following a Suxamethonium, may occur secondary to reduced plasma cholinesterase in the following states or pathological conditions; end stage hepatic failure, acute or chronic renal failure.
An acute transient rise in serum potassium often occurs following the administration of suxamethonium in normal individuals; the magnitude of this rise is of the order of 0.5 mmol/litre. In certain pathological states or conditions this increase in serum potassium following suxamethonium administration may be excessive and cause serious cardiac arrhythmias and cardiac arrest.

For this reason, the use of suxamethonium is contra-indicated in:
Patients recovering from major trauma or severe burns; the period of greatest risk of hyperkalaemia is from about 5 to 70 days after the injury and may be further prolonged if there is delayed healing due to persistent infection.
Patients with neurological deficits involving acute major muscle wasting (upper and/or lower motor neurone lesions); the potential for potassium release occurs within the first 6 months after the acute onset of the neurological deficit and correlates with the degree and extent of muscle paralysis. Patients who have been immobilised for prolonged periods of time may be at similar risk.
Suxamethonium is also contra-indicated in patients with pre-existing hyperkalaemia. In the absence of hyperkalaemia and neuropathy, renal failure is not a contra-indication to the use of a normal single dose of suxamethonium, although multiple or large doses may cause clinically significant rises in serum potassium and should not be used.
Patients with skeletal muscle myopathies e.g. Duchenne muscular dystrophy (increased risk of malignant hyperthermia, ventricular dysrhythmias and cardiac arrest secondary to acute rhabdomyolysis with hyperkalaemia - see above).
Suxamethonium is contra-indicated in patients with cerebral palsy.
Patients with a personal or family history of congenital myotonic diseases such as myotonia congenita and dystrophia myotonica (risk of severe myotonic spasms and rigidity).

WARNINGS AND PRECAUTIONS

Suxamethonium paralyses the respiratory muscles as well as other skeletal muscles but has no effect on consciousness.

Suxamethonium should be administered only by or under supervision of an anaesthetist who is familiar with its actions, characteristics and hazards, who is skilled in the management of artificial respiration and only where there are adequate facilities for immediate endotracheal intubation with the administration of oxygen by intermittent positive pressure ventilation.

Cross-sensitivity

High rates of cross-sensitivity (greater than 50%) between neuromuscular blocking agents have been reported.
Therefore, where possible, before administering suxamethonium, hypersensitivity to other neuromuscular blocking agents should be excluded. Suxamethonium, should only be used when absolutely essential in susceptible patients.
Patients who experience a hypersensitivity reaction under general anaesthesia should be tested subsequently for hypersensitivity to other neuromuscular blockers.
Suxamethonium should not be mixed in the same syringe with any other agent, especially thiopental.
During prolonged administration of suxamethonium, it is recommended that the patient is fully monitored with a peripheral nerve stimulator in order to avoid overdosage.

Hyperkalaemia

Suxamethonium increases serum potassium by 0.5mmol/L in normal individuals. This may be significant with pre-existing elevated serum potassium. Patients with burns or certain neurological conditions may develop severe hyperkalaemia (see section Contraindication). In severe sepsis, the potential for hyperkalaemia may be related to the severity and duration of the infection.

Cholinesterase deficiency

In patients with low levels of plasma cholinesterase or with an abnormal pseudocholinesterase, suxamethonium should be used only with extreme caution and where the benefits of the drug are considered to outweigh the risks.
Suxamethonium is rapidly hydrolysed by plasma cholinesterase which thereby limits the intensity and duration of the neuromuscular blockade. Approximately 0.05% of the population has an inherited cause of reduced cholinesterase activity. Deficiencies of this enzyme result in a prolonged and intensified neuromuscular block. Deficiency may be:

- congenital (genetically determined abnormal plasma cholinesterase)
- physiological (during pregnancy and puerperium)
- acquired (in conjunction with liver disease, chronic debilitating disease,

chronic anaemia, malnutrition, carcinomatosis, pregnancy, uraemia/renal disease, connective tissue disorders, thyroid disorders, severe burns, severe generalised tetanus, tuberculosis and other severe or chronic infections). Recovery from suxamethonium may be delayed in liver disease due to low serum pseudocholinesterase levels, caution is advised.
• iatrogenic (during cardiopulmonary bypass, following plasma exchange/plasmapheresis, and with other drugs – see section Interaction with Other Medicaments).

Prolonged use

With prolonged use of suxamethonium, the characteristic depolarising blockade (Phase I block) may change to one with characteristics of a non-depolarising blockade (Phase II block), leading to prolonged respiratory depression or apnoea. Although the characteristics of a developing Phase II block resemble those of a true non-depolarising block, the former cannot always be fully or permanently reversed by anticholinesterase agents. When a Phase II block is fully established, its effects will then usually be fully reversible with standard doses of neostigmine accompanied by an anticholinergic agent.
Tachyphylaxis occurs after repeated administration of suxamethonium.
Muscle pains are frequently experienced after administration of suxamethonium and most commonly occur in ambulatory patients undergoing short surgical procedures under general anaesthesia. There appears to be no direct connection between the degree of visible muscle fasciculation after Suxamethonium administration and the incidence or severity of pain. The use of small doses of non-depolarising muscle relaxants given minutes before suxamethonium administration has been advocated for the reduction of incidence and severity of suxamethonium-associated muscle pains. This technique may require the use of doses of suxamethonium in excess of 1mg/kg to achieve satisfactory conditions for endotracheal intubation.
It is inadvisable to use suxamethonium in patients with advanced myasthenia gravis, neurological disorders, myotonia or muscular disease.
Although patients with advanced myasthenia gravis are resistant to suxamethonium they develop a state of Phase II block which can result in delayed recovery.

Patients with myasthenic (Eaton-Lambert) syndrome are more sensitive than normal to suxamethonium and the dose should be reduced. Patients in remission from myasthenic Eaton-Lambert syndrome may however demonstrate a normal response to suxamethonium.
Caution should be exercised when using suxamethonium in children, since paediatric patients are more likely to have an undiagnosed myopathy or an unknown predisposition to malignant hyperthermia and rhabdomyolysis, which places them at increased risk of serious adverse events following suxamethonium (see section Contraindications and section Adverse Reactions).

Bradycardia and other cardiac dysrhythmias

In healthy adults, suxamethonium occasionally causes a mild transient slowing of the heart rate on initial administration.
Bradycardias are more commonly observed in children and on repeated administration of suxamethonium in both children and adults. Pre-treatment with intravenous atropine or glycopyrrolate significantly reduces the incidence and severity of suxamethonium-related bradycardia.
In the absence of pre-existing or evoked hyperkalaemia, ventricular arrhythmias are rarely seen following suxamethonium administration. Cardiac arrhythmias can develop in patients receiving digitalis glycosides who are given suxamethonium. Patients taking digitalis-like drugs are however more susceptible to such arrhythmias. The action of suxamethonium on the heart may cause changes in cardiac rhythm including cardiac arrest.

Raised intra-ocular pressure (IOP)

Suxamethonium causes a transient increase in intraocular pressure and should not be used in the presence of penetrating eye injury except where the potential benefits outweigh the injury to the eye.
Use in other conditions
This agent should be used with caution in ill and cachectic patients, in patients with acid-base disturbances or electrolyte imbalance, parenchymatous liver disease, obstructive jaundice, carcinomatosis, in those in contact with certain insecticides, e.g. organophosphorus compounds and in those receiving therapeutic radiation. Suxamethonium should be used with caution in patients with fractures or muscle spasms because the initial muscle fasciculations may cause additional trauma. Muscarinic effects of this compound e.g. increased bronchial and salivary secretions may be prevented by atropine.
When this agent is given as an infusion, this should be monitored with care to avoid overdosing.
Suxamethonium has no direct effect on the myocardium, but by stimulation of both autonomic ganglia and muscarinic receptors suxamethonium may cause changes in cardiac rhythm, including cardiac arrest.

INTERACTION WITH OTHER MEDICAMENT

Certain drugs or chemicals are known to reduce normal plasma cholinesterase activity and may therefore prolong the neuromuscular blocking effects of suxamethonium. These include:

Antibacterials:

Enhanced effects of suxamethonium with aminoglycosides, clindamycin, polymyxins and vancomycin and piperacillin.

Antimalarials:

Quinine - effects of suxamethonium possibly enhanced.

Antipsychotics:

Enhanced effects of suxamethonium with promazine, promethazine, chlorpromazine, phenelzine and lithium carbonate.

General anaesthetic agents:

Propofol (increased risk of myocardial depression and bradycardia), volatile liquid gas (enhanced effects of suxamethonium) and ketamine and propanid (possible prolonged block).

Analgesics:

Enhanced effects of suxamethonium with morphine, morphine antagonists, pethidine and pancuronium.

Anti-arrhythmics:

Lidocaine (or lignocaine) (enhanced and prolonged neuromuscular blockade), quinidine, procainamide and verapamil and beta-blockers (enhanced and prolonged neuromuscular blockade)

Local anaesthetics:

Enhanced effects of suxamethonium with procaine, cocaine, chlorprocaine and lidocaine (see above).

Cardiac glycosides:

Possible increased risk of bradycardia and other dysrhythmias, including ventricular dysrhythmias and cardiac arrest and more susceptible to the effects of suxamethonium exacerbated by hyperkalaemia.

Cytotoxics:

Enhanced effects of suxamethonium with cyclophosphamide, thiotepa, other alkylating agents (chloroethamine: tretamine) and triethylene-melamine.

Immunomodulators:

Azathioprine (prolonged neuromuscular blockade).

Magnesium:

Parenteral magnesium (enhanced neuromuscular blockade).

Metoclopramide:

Enhanced effects of suxamethonium.

Parasympathetics:

Enhanced effects of suxamethonium with donepezil, edrophonium, galantamine, neostigmine, physostigmine, pyridostigmine and rivastigmine and tacrine hydrochloride.

Sympathomimetics (beta agonists):

Bambuterol and terbutaline (enhanced effects of suxamethonium)

Anti-histamines:

Diphenhydramine (enhanced effects of suxamethonium).

Volatile inhalational anaesthetic agents:

Halothane, enflurane, desflurane, isoflurane, diethylether and methoxyflurane have little effect on the Phase I block of Suxamethonium Chloride injection but will accelerate the onset and enhance the intensity of a Phase II suxamethonium-induced block.

Drugs which are known to reduce normal plasma cholinesterase:

In addition to the drugs listed above, certain other drugs and chemicals are known to reduce normal plasma cholinesterase activity and therefore may prolong the neuromuscular effect of suxamethonium. These include organophosphorous insecticides and metrifphonate, ecothiopate eye drops (prolonged apnoea after suxamethonium has occurred), trimethaphan and selective serotonin reuptake inhibitors (SSRI).

The following have potentially adverse effects on plasma cholinesterase activity: aprotinin, oestrogens and oral contraceptives, oxytocin and high-dose steroids.

Liver disease, cancer, pregnancy, dehydration, electrolyte imbalances and overdosage (due to excessive production of succinylmonocholine) may also prolong the action of suxamethonium.

INCOMPATIBILITIES

This medicinal product must not be mixed in the same syringe with any other agent especially thiopentone.

USE IN PREGNANCY AND LACTATION

Pregnancy

Suxamethonium has no direct action on the uterus or other smooth muscle structures. In normal therapeutic doses it does not cross the placental barrier in sufficient amounts to affect the respiration of the infant.

The benefits of the use of suxamethonium as part of a rapid sequence induction for general anaesthesia normally outweigh the possible risk to the foetus. Plasma cholinesterase levels fall during the first trimester of pregnancy to about 70 to 80% of their pre-pregnancy values; a further fall to about 60 to 70% of the pre-pregnancy levels occurs within 2 to 4 days after delivery. Plasma cholinesterase levels then increase to reach normal over the next 6 weeks. Consequently, a high proportion of pregnant and puerperal patients may exhibit mildly prolonged neuromuscular blockade following Suxamethonium Injection. Suxamethonium Chloride Injection should not be used unless clearly necessary.

Breast-Feeding

It is not known whether suxamethonium is excreted in breast milk, therefore, caution should be exercised following administration of suxamethonium to nursing mothers.

ADVERSE EFFECTS

The frequency categories assigned to the adverse reactions are estimates. For most reactions, the frequency was determined from published data and the background incidence was not considered when determining the frequency groups. Adverse reactions are listed below by system organ class and frequency. Estimated frequencies were determined from published data.

Frequencies are defined as follows: very common; common, uncommon; rare; very rare.

System organ class	Frequency	Undesirable effects
Immune system disorders	Very Rare	Anaphylactic reactions
Eye disorders	Common	Increased intraocular pressure
Cardiac disorders	Common Rare	Bradycardia, tachycardia Arrhythmias (including ventricular arrhythmias), cardiac arrest ¹
Vascular disorders	Common	Skin flushing Hypertension and hypotension have also been reported.
Respiratory, thoracic and mediastinal disorders	Rare	Bronchospasm, prolonged respiratory depression ² , apnoea.
Gastrointestinal disorders	Very Common	Increased intragastric pressure Excessive salivation has also been reported
Skin and subcutaneous tissue disorders	Common	Rash
Musculoskeletal and connective tissue disorder	Very common Common Rare	Muscle fasciculation, post-operative muscle pains Myoglobinaemia ³ , myoglobinuria ³ Trismus
General disorders and administration site conditions	Vary rare	Malignant hyperthermia
Investigations	Common	Transient blood potassium increase

¹ There are case reports of hyperkalaemia-related cardiac arrests following the administration of suxamethonium to patients with congenital cerebral palsy, tetanus, Duchenne muscular dystrophy, and closed head injury. Such events have also been reported rarely in children with hitherto undiagnosed muscular disorders.

² Individuals with decreased plasma cholinesterase activity exhibit a prolonged response to suxamethonium. Approximately 0.05% of the population has an

inherited cause of reduced cholinesterase activity (Please refer to section Special Warnings and Precautions for Use)

³ Rhabdomyolysis has also been reported (see section Contraindication and Special Warnings and Precautions for Use)

SYMPTOMS AND TREATMENT OF OVERDOSE

Symptoms:

Apnoea and prolonged muscle paralysis are the main and serious effects of overdosage.

Treatment:

It is essential to maintain the airway and to ensure adequate ventilation until spontaneous respiration occurs.

Neostigmine and other anticholinesterase drugs are not antidotes to suxamethonium but would normally intensify the depolarisation effect. However, in some cases when the action of suxamethonium is prolonged, the characteristic depolarising (Phase I) block may change to one with characteristics of a non-depolarising (Phase II) block.

The decision to use neostigmine to reverse a Phase II suxamethonium-induced block depends on the judgement of the clinician in the individual case. Valuable information in regard to this decision will be gained by monitoring neuromuscular function.

To investigate this possibility, the short-acting anticholinesterase drug, edrophonium, may be given intravenously. If an obvious improvement is maintained for several minutes, neostigmine may be given with atropine. Subsequently, the patient should be observed carefully and if apnoea recurs, a further dose of neostigmine is indicated.

EFFECTS ON ABILITY TO DRIVE AND USE OF MACHINE

Suxamethonium chloride has a major influence on the ability of an individual to drive or operate machinery.

This precaution is not relevant to the use of Suxamethonium Injection. Suxamethonium will always be used in combination with a general anaesthetic and therefore the usual precautions relating to the performance of tasks following general anaesthesia apply.

INSTRUCTION FOR USE/ HANDLING

If only part used, discard the remaining solution immediately after first use.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

STORAGE CONDITIONS

Store in a refrigerator 2-8°C.

Protect from light.

Do not freeze.

Retain in the container until time of use.

The solution should be colourless solution when diluted. The solution will remain stable for 24 hours at both temperature 2-8°C and below 30°C after diluted with the compatible solutions as below:

Infusion solution	Period of stability (hours)
0.9% Sodium chloride Solution	24 hours
5% Dextrose Solution	

Use once and discard any remaining solution.

SHELF LIFE:

The injections can be used within 18 months from the date of manufacture if kept as recommended.

The solution should be colourless solution when diluted. The solution will remain stable for 24 hours at both temperature 2-8°C and below 30°C after diluted with the compatible solutions as below:

Infusion solution	Period of stability (hours)
0.9% Sodium chloride Solution	24 hours
5% Dextrose Solution	

DOSAGE FORM AND PACKAGING AVAILABLE

10 x 2mL ampoule (amber glass ampoule) of Pharmaniaga Suxamethonium Chloride 100mg/2mL Solution for Injection.

10 ampoules per unit PVC tray were packed in a paper carton.

PRODUCT REGISTRATION HOLDER/ MANUFACTURER

PHARMANIAGA LIFESCIENCE SDN BHD (198201002939)

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